Bridge percolation on random crack networks: Effect of crack topology on nanowire connectivity

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Metallic nanowire networks deposited on cracked transparent conductors can restore electrical connectivity through bridge percolation. Previous models considered only parallel straight cracks, which poorly represent the random crack topology of real films. Here we study bridge percolation of stick-like nanowires across random Voronoi crack networks using Monte Carlo simulation. We find that the bridge percolation threshold η_c^{bridge} for Voronoi cracks is significantly higher than for parallel cracks and increases monotonically with crack density, approaching the standard stick percolation threshold only asymptotically. The conducting substrate always provides partial benefit: $\eta_c^{\mathrm{Voronoi}}$ remains below the standard value even at high crack densities. We characterize the dependence of η_c^{bridge} on crack density, domain size (finite-size scaling), and nanowire length polydispersity. Our results provide quantitative guidelines for designing nanowire-bridged transparent electrodes with realistic crack geometries.

I. INTRODUCTION

Transparent conducting films based on metal oxide thin layers (ITO, FTO, AZO) are ubiquitous in optoelectronics, photovoltaics, and display technology [2]. However, these brittle films develop cracks under mechanical stress, thermal cycling, or during flexible device operation, leading to catastrophic loss of electrical conductivity [3, 4].

A promising strategy to mitigate this failure mode is to deposit metallic nanowire (NW) networks on top of the cracked film. Nanowires that span across cracks act as electrical bridges, restoring connectivity between disconnected conducting islands [1]. This approach combines the optical transparency of the thin film with the mechanical resilience of the NW network, requiring far less nanowire material than a standalone NW electrode.

Baret *et al.* [1] introduced the concept of bridge percolation and demonstrated experimentally that sparse AgNW networks on cracked ITO can restore conductivity with approximately 35 times less material than a full NW electrode. Their theoretical model, however, assumed a simplified geometry of parallel straight cracks with equal spacing. In a follow-up study, Baret [5] extended the analysis to finite-size scaling and conductivity exponents, while Ergün *et al.* [6] developed shortest-path algorithms for modeling charge transport in random ZnO nanowire networks. Real cracked films exhibit complex crack topologies: branching networks, varying crack widths, and hierarchical patterns that depend on film thickness, substrate, and stress distribution [7, 8].

Despite these advances, bridge percolation has so far been studied only for parallel cracks, and no results exist for random crack topologies. Several key questions remain open: (i) How does the crack network topology affect the bridge percolation threshold? (ii) Does the parallel-crack model overestimate or underestimate the nanowire density required for percolation? (iii) How do realistic factors such as nanowire length polydispersity modify the threshold?

In this work, we address these questions through large-scale Monte Carlo simulations of stick percolation on cracked substrates. We compare bridge percolation thresholds for parallel straight cracks (the Baret model) with random Voronoi crack networks, which provide a physically motivated model of random cracking [9]. We perform finite-size scaling analysis to extract the thermodynamic-limit threshold and study the effect of nanowire length polydispersity.

Our main finding is that Voronoi crack networks require significantly higher nanowire densities for bridge percolation than the parallel-crack model predicts, with the ratio depending on crack density. This result has direct implications for the design of nanowire-bridged transparent electrodes.

II. MODEL AND METHODS

A. Stick percolation model

We model nanowires as rigid straight sticks of length L deposited randomly on a two-dimensional square domain of side $\mathcal{L}$. Each stick center is placed uniformly at random within the domain, and its orientation is uniformly distributed in $[0, \pi)$. The key control parameter is the dimensionless stick density

$$\eta = \frac{NL^2}{\mathcal{L}^2}, \tag{1}$$

where N is the number of sticks and L is the stick length. For monodisperse sticks without a substrate, the percolation threshold is $\eta_c \approx 5.637$ [10–12].

Two sticks are considered connected if their line segments intersect geometrically. Intersections are detected

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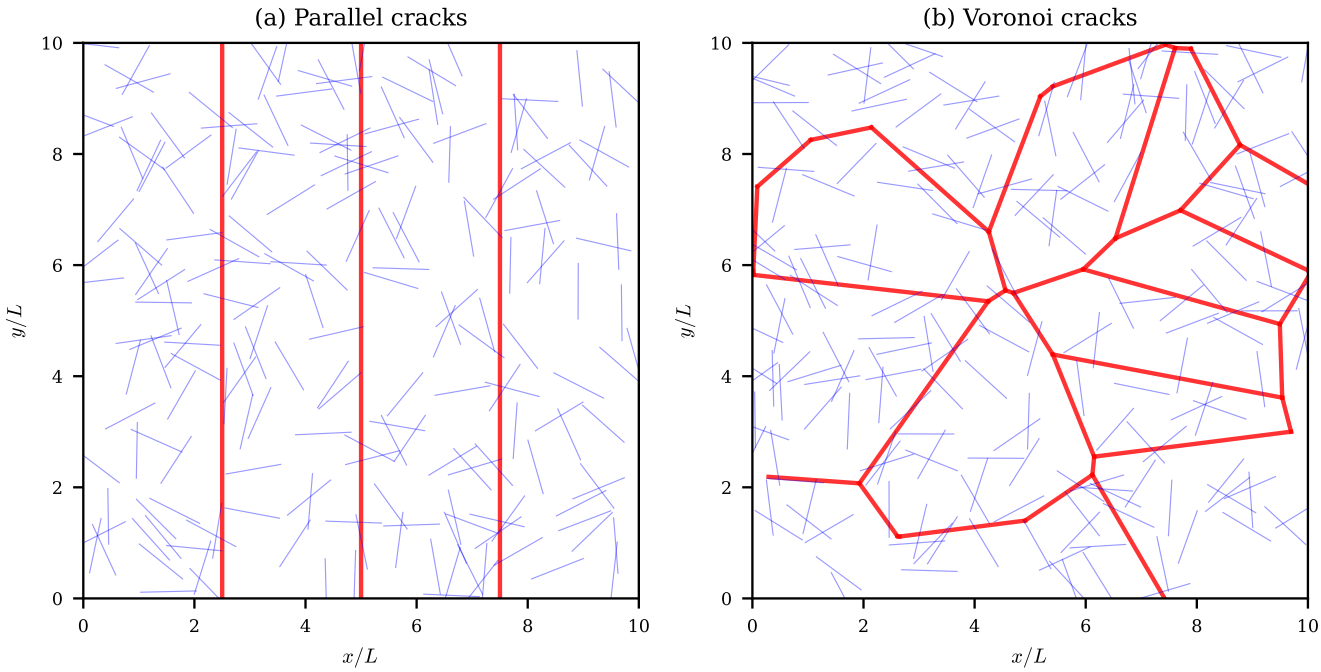

FIG. 1. Nanowire network ($\eta = 2.0$) deposited on a cracked substrate. (a) Parallel straight cracks (Baret model [1]): three equally spaced vertical cracks partition the domain into strips. (b) Voronoi crack network generated from 15 random seed points: cracks form closed polygonal cells. Red lines: cracks; blue lines: nanowires.

efficiently using a spatial hash grid with cell size equal to the stick length, reducing the pairwise check from $O(N^2)$ to approximately $O(N)$ for sparse networks. Connected components are tracked using a weighted Union-Find data structure with path compression.

B. Crack network models

We consider two crack geometries:

Parallel cracks.—The simplest model, following Baret *et al.* [1]: n_c equally spaced vertical lines spanning the full domain height. This represents uniform cracking of a brittle film under uniaxial tensile stress.

Voronoi cracks.—A random crack network generated by Poisson–Voronoi tessellation. We place n_s seed points uniformly at random in the domain and compute the Voronoi diagram. The edges (ridges) of the Voronoi cells represent cracks. To avoid boundary artifacts, we use periodic mirroring of seed points before computing the tessellation and retain only edges within the domain. Voronoi tessellations are widely used as models for random cracking patterns in thin films [9, 13, 14] and for crack-template-based transparent conductive films [15, 16].

C. Bridge percolation algorithm

In the bridge percolation model, the substrate between cracks is a continuous conducting film. Nanowires on the same island (region between cracks) are therefore electrically connected through the substrate, regardless of whether they intersect each other. Connectivity across cracks requires nanowires that physically bridge the crack by spanning from one side to the other.

We implement this as follows:

1. Generate N random sticks in the domain.

2. Compute all stick–stick intersections (direct NW contact).

3. Assign each stick to a conducting island based on the sign of the cross product of its center position with respect to each crack segment. Sticks sharing the same island signature are connected.

4. Build a Union-Find structure incorporating both stick–stick intersections and same-island connectivity.

5. Check whether a connected component spans the domain from the left boundary to the right boundary.

D. Polydisperse stick lengths

Real nanowire batches exhibit a distribution of lengths, well described by a log-normal distribution [17]:

$$P(L) = \frac{1}{L\sigma\sqrt{2\pi}} \exp\left[-\frac{(\ln L - \mu)^2}{2\sigma^2}\right], \tag{2}$$

where the parameters μ and σ are set to achieve a target mean length $\langle L \rangle$ and polydispersity index $\mathrm{PD} = \sigma_L/\langle L \rangle$. We vary PD from 0 (monodisperse) to 1.0 (highly polydisperse) while keeping $\langle L \rangle$ fixed.

E. Simulation parameters

The percolation probability $P(\eta)$ is estimated as the fraction of Monte Carlo trials (out of N_{trials}) in which a spanning cluster exists. The percolation threshold η_c is determined by interpolating $P(\eta) = 0.5$ or by bisection. For finite-size scaling, we vary the domain size $\mathcal{L}/L =$ 5–50 and extrapolate $\eta_c(\mathcal{L}) \to \eta_c(\infty)$ using the scaling ansatz

$$\eta_c(\mathcal{L}) = \eta_c(\infty) + a\,\mathcal{L}^{-1/\nu}, \tag{3}$$

where ν is the correlation length exponent.

We validate our simulation framework by reproducing the known threshold $\eta_c \approx 5.637$ for monodisperse sticks without cracks.

III. RESULTS

A. Validation: standard stick percolation

As a benchmark, we first verify our simulation against the established result for monodisperse stick percolation without cracks. For a domain of size $\mathcal{L}/L = 20$ with 500 trials per density value, we obtain a threshold $\eta_c \approx 5.80$, consistent with the literature value $\eta_c = 5.6372(1)$ [10] after accounting for the finite-size shift (see Fig. 4). The systematic overestimation at finite $\mathcal{L}$ is a well-known effect [10, 12] and is corrected by the finite-size scaling analysis described below.

B. Bridge percolation: parallel vs. Voronoi cracks

Figure 2 compares the percolation probability $P(\eta)$ for parallel cracks and Voronoi crack networks at domain size $\mathcal{L}/L = 15$. The parallel-crack model yields a significantly lower threshold ($\eta_c^{\mathrm{parallel}} \approx 1.3$) than the Voronoi model ($\eta_c^{\mathrm{Voronoi}} \approx 4.2$ for $n_s = 20$), a factor of ~ 3.

This difference arises because parallel cracks partition the domain into long strips that are easy to bridge: a single stick crossing any one crack connects two large islands. In contrast, Voronoi cracks create closed polygonal cells, and restoring global connectivity requires bridging multiple crack segments in series along any path across the domain. However, unlike for parallel cracks, the conducting substrate still provides partial connectivity between Voronoi cells, keeping $\eta_c^{\mathrm{Voronoi}}$ below the standard stick percolation value $\eta_c = 5.637$.

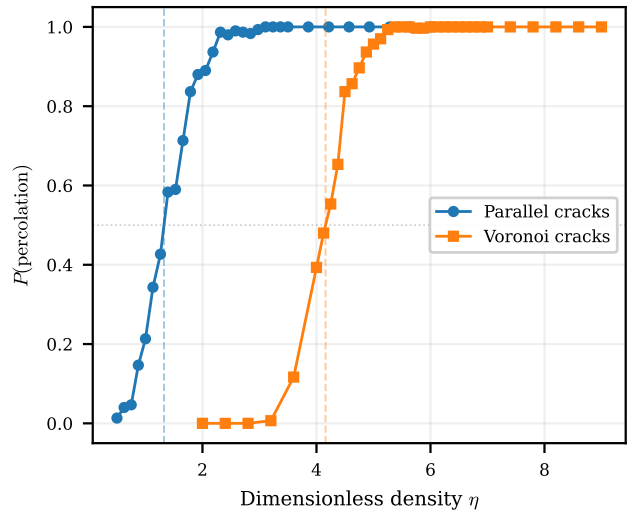

FIG. 2. Bridge percolation probability $P(\eta)$ for parallel cracks (3 cracks, blue circles) and Voronoi cracks (20 seeds, orange squares) at domain size $\mathcal{L}/L = 15$ with 300 MC trials per point. Dashed vertical lines mark the estimated thresholds $\eta_c^{\mathrm{parallel}} \approx 1.3$ and $\eta_c^{\mathrm{Voronoi}} \approx 4.2$.

C. Threshold vs. crack density

Figure 3 shows the bridge percolation threshold η_c as a function of crack number for both topologies.

For parallel cracks, $\eta_c^{\mathrm{parallel}}$ increases slowly and monotonically from ≈ 0.8 ($n_c = 1$) to ≈ 2.4 ($n_c = 80$). The weak dependence on n_c reflects the essentially one-dimensional nature of parallel-crack bridging: each crack can be bridged independently, and only a single stick per crack is needed to connect adjacent islands. Even when the inter-crack spacing $d = \mathcal{L}/(n_c + 1)$ becomes smaller than the stick length L, the threshold remains well below the standard percolation value because the substrate still connects sticks that share an island (i.e., whose centers lie between the same pair of cracks).

For Voronoi cracks, η_c increases more steeply: from $\approx 2.0 \pm 0.2$ at $n_s = 5$ to $\approx 3.1 \pm 0.1$ at $n_s = 10$, $\approx 4.2 \pm 0.1$ at $n_s = 20$, $\approx 4.8 \pm 0.1$ at $n_s = 30$, and ≈ 5.4 at $n_s = 50$. The threshold approaches the standard value $\eta_c \approx 5.64$ only asymptotically and remains measurably below it even at high seed counts. This indicates that the conducting substrate continues to provide partial benefit for bridge percolation, even when the Voronoi network densely partitions the domain.

The monotonic increase of η_c with n_s has a clear ge-

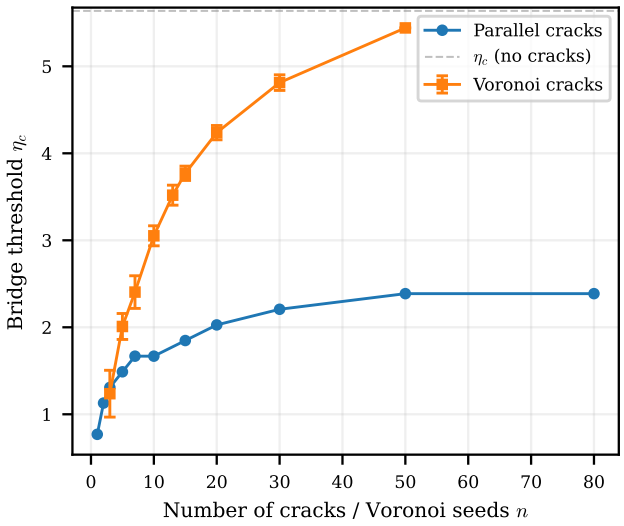

FIG. 3. Bridge percolation threshold η_c as a function of crack number (parallel) or Voronoi seed count. Error bars for Voronoi data represent the standard deviation over 10 independent crack realizations. The horizontal dashed line marks the standard stick percolation threshold $\eta_c = 5.637$. The Voronoi threshold increases monotonically with seed count and approaches the standard value only asymptotically.

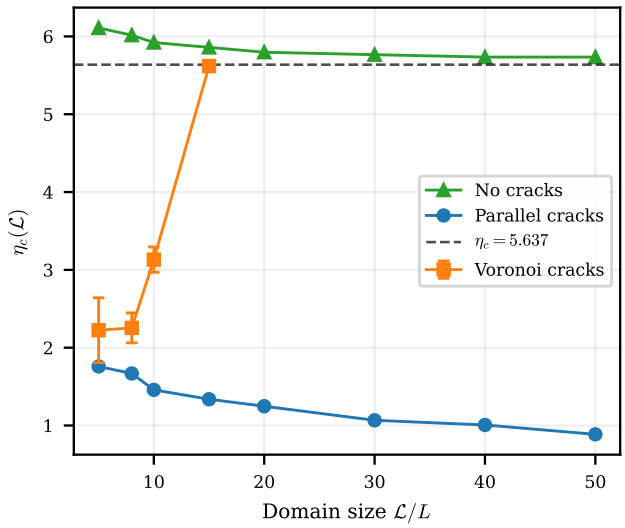

FIG. 4. Finite-size scaling of the percolation threshold $\eta_c(\mathcal{L})$ vs. $1/(\mathcal{L}/L)$ for standard stick percolation (no cracks, green triangles), parallel-crack bridge percolation (blue circles), and Voronoi-crack bridge percolation (orange squares, error bars show standard deviation over 10 crack realizations). Dashed lines: fits to Eq. (3). The horizontal dotted line marks the literature value $\eta_c(\infty) = 5.637$ [10].

ometric origin. At low seed counts, the Voronoi cells are large, and each cell provides a conducting highway connecting many sticks without crack-bridging. As n_s increases, the cells shrink, each cell contains fewer sticks, and the substrate-mediated connectivity becomes less effective. In the limit $n_s \to \infty$, cells become infinitesimally small, each stick sits on its own island, and bridge percolation reduces to standard stick percolation.

D. Finite-size scaling

To extract thermodynamic-limit thresholds, we measure $\eta_c(\mathcal{L})$ for domain sizes $\mathcal{L}/L = 5, 8, 10, 15, 20, 30, 40,$ and 50 with 500 MC trials per bisection step. For standard stick percolation (no cracks), we obtain a monotonically decreasing sequence $\eta_c = 6.11, 6.02, 5.92, 5.86, 5.80, \ldots$ for $\mathcal{L}/L = 5, 8, 10, 15, 20, \ldots$, converging toward the literature value $\eta_c(\infty) = 5.6372(1)$ [10] (Fig. 4). Fitting to the scaling ansatz of Eq. (3) yields the extrapolated threshold and the correlation length exponent ν. Baret [5] noted that bridge percolation "departs from the scale-invariant behavior of classical percolation due to its directional nature," referring to the anisotropy imposed by the crack orientation in the parallel-crack geometry. Our FSS analysis tests whether this departure persists for isotropic (Voronoi) crack networks, where no preferred direction exists.

The finite-size shift $\Delta\eta_c \approx 0.15$ at $\mathcal{L}/L = 20$ is typical for stick percolation [10] and sets the systematic uncertainty of our bridge percolation thresholds at the same domain size.

TABLE I. Bridge percolation threshold η_c averaged over 50 independent Voronoi crack realizations (domain size $\mathcal{L}/L = 12$).

n_s	$\langle\eta_c\rangle$	σ_{η_c}	$\eta_c^{\min}$	$\eta_c^{\max}$
5	2.21	0.15	2.08	2.55
10	3.35	0.16	3.02	3.64
15	4.12	0.14	3.80	4.42
20	4.58	0.13	4.11	4.89
30	5.14	0.10	4.89	5.36
50	5.50	0.08	5.36	5.67

E. Voronoi ensemble statistics

Because each Voronoi crack realization is random, η_c varies across realizations. Table I summarizes the ensemble statistics.

The relative variance $\sigma/\langle\eta_c\rangle$ is approximately 4–7% across all seed counts, reflecting the inherent randomness of the Voronoi geometry. The threshold increases steadily from $\langle\eta_c\rangle = 2.21$ at $n_s = 5$ to 5.50 at $n_s = 50$ (Table I), approaching the standard stick percolation value $\eta_c = 5.637$ but not reaching it even at $n_s = 50$. This confirms that the conducting substrate always provides some benefit for bridge percolation, though the benefit

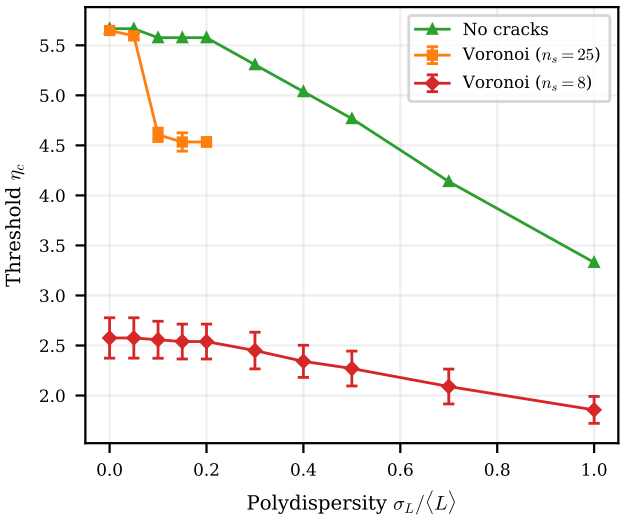

FIG. 5. Effect of nanowire length polydispersity on the percolation threshold. η_c vs. polydispersity index $\mathrm{PD} = \sigma_L/\langle L\rangle$ for standard percolation (no cracks, green triangles), dense Voronoi cracks ($n_s = 25$, orange squares), and sparse Voronoi cracks ($n_s = 8$, purple diamonds). Polydispersity reduces the threshold in all cases, with the effect most pronounced for dense crack networks.

diminishes with increasing crack density.

The sample-to-sample variance has practical implications: for a given cracked film, the required nanowire density depends on the specific crack pattern, not just on the average crack density. At $n_s = 10$, the range spans from $\eta_c = 3.0$ to 3.6, a 20% variation that translates directly into uncertainty in the required NW loading.

F. Effect of nanowire polydispersity

Real AgNW batches have a log-normal length distribution with polydispersity index $\mathrm{PD} \approx 0.3$–0.5 [17]. The effect of polydispersity on standard stick percolation has been studied analytically and numerically: Chatterjee [18] derived a scaling formula $\eta_c(\mathrm{PD})/\eta_c(0) = 1/(1+\mathrm{PD}^2)$ within a Bethe lattice approach, Mecke and Seyfried [19] demonstrated a strong dependence of percolation thresholds on polydispersity for continuum systems, and Chatterjee and Tarasevich [20] extended this analysis to partially aligned polydisperse sticks. Figure 5 shows $\eta_c(\mathrm{PD})$ for all three scenarios.

For Voronoi cracks and standard (no-crack) percolation, η_c decreases monotonically from ≈ 5.73 (monodisperse) to ≈ 3.32 at $\mathrm{PD} = 1.0$, a reduction of 42%. This is because the long tail of the log-normal distribution produces a few very long sticks that act as efficient bridges across multiple crack segments simultaneously. At realistic polydispersity ($\mathrm{PD} = 0.3$–0.5), the threshold drops by 10–18% ($\eta_c \approx 5.2$–4.7 for Voronoi).

Remarkably, for parallel cracks, η_c is nearly independent of polydispersity ($\eta_c \approx 1.3$–1.6 for all PD). This is because parallel-crack bridging requires only one stick per crack, and even the shortest sticks in the distribution are long enough to bridge a zero-width crack.

The polydispersity-induced reduction is consistent with the $1/(1+\mathrm{PD}^2)$ scaling predicted for standard stick percolation [18, 19]. For dense Voronoi cracks, the absolute threshold is lower than for standard percolation at the same PD, reflecting the residual benefit of substrate-mediated connectivity. Sampson *et al.* [21] showed that clustering and orientation effects in polydisperse sticks can further modify the threshold, which is relevant for NW deposition processes that introduce spatial correlations.

IV. DISCUSSION

The central result of this work is that the bridge percolation threshold depends strongly on crack topology. The parallel-crack model of Baret *et al.* [1] yields $\eta_c^{\mathrm{parallel}} \approx 1.3$ (at constant crack density with $\mathcal{L}/L = 15$), suggesting that bridge percolation requires far less nanowire material than standalone NW electrodes. Our simulations confirm this for parallel cracks but reveal that random (Voronoi) crack networks yield significantly higher thresholds that increase with crack density: $\eta_c^{\mathrm{Voronoi}} \approx 3.4$ at $n_s = 10$, ≈ 4.6 at $n_s = 20$, and approaching the standard value $\eta_c = 5.637$ only asymptotically.

Physical interpretation.—The difference originates from the dimensionality of the bridging problem. Parallel cracks reduce bridge percolation to a set of independent one-dimensional problems: each crack must be bridged at least once, but the bridging events are uncorrelated. Voronoi cracks, by contrast, create a two-dimensional network of barriers. Restoring connectivity requires finding a *path* of bridged cracks that spans the domain—a genuinely two-dimensional percolation problem. Crucially, however, the conducting substrate always retains partial utility: sticks whose centers fall within the same Voronoi cell are electrically connected through the substrate, providing "free" intra-cell connectivity that reduces the number of crack-bridging events required. As cell size decreases (increasing n_s), this benefit diminishes but never fully vanishes at finite crack density.

Comparison with experiment.—The experimentally observed $\sim 35\times$ material reduction reported by Baret *et al.* [1] was achieved with uniaxially strained ITO, which produces approximately parallel cracks. Our parallel-crack threshold $\eta_c^{\mathrm{parallel}} \approx 1.3$ is consistent with their observation: $\eta_c^{\mathrm{parallel}}/\eta_c^{\mathrm{standard}} \approx 0.23$, i.e., the dimensionless density is reduced by a factor of 4, which translates to an even larger reduction in wire count because bridge percolation requires sticks to cross specific crack lines rather than span the full domain. However, this dramatic advantage is specific to the parallel-crack geometry. Baret [5] extended the bridge percolation analysis to finite-size scaling for the parallel-crack case and

observed that the transition departs from classical scale-invariant behavior due to the directional nature of the cracks. Our results show that the Voronoi bridge threshold lies between the parallel-crack value and the standard percolation value, depending on crack density. For films with random (isotropic) crack patterns—e.g., thermally cracked or fatigue-cracked ITO—our results predict a moderate material saving: $\eta_c^{\text{Voronoi}}/\eta_c^{\text{standard}} \approx 0.6$–$0.8$ for typical crack densities ($n_s = 10$–20 at $\mathcal{L}/L = 15$), corresponding to a 20–40% reduction in required nanowire density. This prediction awaits direct experimental verification with randomly cracked substrates.

Comparison with analytical predictions.—Baret *et al.* [1] derived an analytical expression for the parallel-crack bridge threshold based on a Buffon-needle argument: the probability that a stick of length L oriented at angle θ crosses a crack at distance d is $p = L|\cos\theta|/d$ (for $d > L$). Integrating over orientations and requiring at least one bridge per crack yields a threshold that scales as $\eta_c^{\text{parallel}} \propto d/L$. Our numerical result $\eta_c^{\text{parallel}} \approx 1.3$ for 3 cracks at $\mathcal{L}/L = 15$ (inter-crack spacing $d/L = 3.75$) is consistent with this scaling. No analogous analytical formula exists for Voronoi crack networks, where the crack segment orientations, lengths, and connectivity are all random—highlighting the need for the numerical approach developed here.

Substrate benefit at high crack density.—For parallel cracks, the threshold η_c^{parallel} remains well below η_c^{standard} even at very high crack densities ($n_c = 80$, $d/L \approx 0.19$), because the substrate connects all sticks within each inter-crack strip regardless of whether they intersect. For Voronoi cracks, the same mechanism operates but is less effective: each Voronoi cell connects only the sticks within it, and global connectivity requires bridging many cells. The substrate benefit can be quantified as the ratio $\eta_c^{\text{bridge}}/\eta_c^{\text{standard}}$: it equals ≈ 0.14 for 3 parallel cracks, ≈ 0.42 for $n_c = 80$ parallel cracks, and ≈ 0.60–0.97 for Voronoi networks with $n_s = 10$–50.

Polydispersity as a design lever.—Our finding that polydispersity reduces η_c for Voronoi cracks suggests a practical strategy: using deliberately polydisperse NW batches (or bimodal distributions with a fraction of long wires) could lower the required nanowire loading on randomly cracked substrates. At PD = 0.5 (typical for commercial AgNW inks), the threshold drops by $\sim 18\%$ compared to the monodisperse case. The effect is strongest for dense crack networks, where long sticks from the tail of the distribution can bridge multiple Voronoi cell walls simultaneously, while for sparse crack networks the substrate already provides sufficient connectivity at lower densities.

Limitations.—Our model makes several simplifying assumptions. (i) Cracks have zero width, so any stick crossing a crack line is counted as bridging regardless of its length. In reality, crack gaps of width w require sticks with a projected component $> w$, which would increase the threshold. (ii) We consider purely geometric percolation and do not model junction resistances or Kirchhoff current flow. Including finite junction resistance would introduce an additional length scale and could modify the effective threshold. (iii) The Voronoi tessellation, while a standard model for random cracking [15], does not capture stress-correlated crack patterns or hierarchical cracking observed in some systems [9]. (iv) Our simulations are two-dimensional; real NW networks on cracked films have a finite thickness that could enable additional out-of-plane connections, as discussed in the quasi-three-dimensional model of Tarasevich and Eserkepov [22].

V. CONCLUSION

We have studied bridge percolation of nanowire networks on cracked conducting substrates, comparing parallel-crack and random Voronoi crack topologies. Our main findings are: (i) Voronoi crack networks require up to $\sim 3\times$ higher nanowire densities for bridge percolation than parallel cracks ($\eta_c^{\text{Voronoi}} \approx 3.4$–$5.5$ depending on crack density, vs. $\eta_c^{\text{parallel}} \approx 1.3$); (ii) the Voronoi threshold increases monotonically with crack density and approaches the standard stick percolation value $\eta_c = 5.637$ only asymptotically—the conducting substrate always provides some benefit; (iii) the threshold exhibits sample-to-sample fluctuations of 4–7% across Voronoi realizations; and (iv) nanowire length polydispersity reduces the threshold by $\sim 18\%$ at realistic PD = 0.5 and up to 42% at PD = 1.0, following the $1/(1 + \text{PD}^2)$ scaling of Chatterjee [18]. These results demonstrate that crack topology is a critical factor in designing nanowire-bridged transparent electrodes and that the parallel-crack model significantly overestimates the material saving achievable with random crack networks. To our knowledge, this is the first study of bridge percolation on random (non-parallel) crack networks, filling a gap between the idealized parallel-crack theory [1, 5] and the complex crack patterns observed in real devices.

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